FlowA: Geometry-Aware State-Intervention Reformulation for Frozen Flow-Matching Checkpoints

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***Abstract*—We introduce FlowA, a geometry-aware, training-free re-inference layer for frozen flow-matching checkpoints. We reformulate inference on both sides of the sampling relation while keeping the learned velocity field fixed: adapters express restart states in checkpoint-native geometry, typed policies blend state and fresh input, and a scheduler allocates correction capacity by round and record difficulty. The resulting frozen ODE is evaluated as a controlled composite map, with each transition preserved in a hash-linked state bundle. We establish conditional fixed-point convergence and an exact squared-error identity for the restart blend, then connect these properties to mechanism-isolating evidence across eleven protein, molecular, toy, and image cells. Native-state alignment improves the Kanzi reconstruction path; restart glue carries the LineageFlow gain in the fixed-NFE round-count study; and difficulty-aware capacity concentrates foldability gains on hard protein records across adapters. Image cells further trace how the intervention responds to round, schedule, and checkpoint regime. FlowA turns mature frozen checkpoints into inspectable, geometry-aware inference systems without retraining.**

***Index Terms*—Flow matching, geometry-aware inference, frozen checkpoints, state intervention, typed contracts.**

I. Introduction

Frozen flow-matching checkpoints generate samples by integrating a learned velocity field once from a prior state. The reformulation acts in two linked places: it expresses the next initial condition as a geometry-aware restart blend and turns the one-shot endpoint evaluation into an iterated composite map. The ODE and learned velocity field remain fixed, while the state transition becomes explicit, typed, and measurable. A four-port contract exposes schedule, policy, merge, and restart rules across protein, molecule, toy, and image adapters; a per-round state bundle and hash-chained ledger preserve each decision for replay.

Our contributions are (i) an inference-dynamics reformulation with a conditional contraction theorem; (ii) an exact restart-blend identity that provides a per-round state diagnostic; (iii) typed integration with five checkpoint adapters and zero edits to vendored code; and (iv) an eleven-cell evaluation organized around mechanism isolation and operating regimes. The evidence includes four strong main-cell results, one tie, and six partial or supplementary cells. R1 is single-seed; R3's paired per-record evidence is measured at $N = 200$ and the larger paired result is projected; R6's overall tier-aware result is counterfactual. The 2.5–10× cross-budget NFE range is an extrapolation from published trajectories, not a measured wall-clock speedup. A direct Tier-1 measurement yields 1.0× NFE speedup (`wave73_phase2_tier1_speedup.json`). Matched-NFE image quality depends on round count and checkpoint setting (Secs. 7.6 and 7.8). Four prespecified Bonferroni families and 30 byte-stable D.4 regression vectors support the evidence chain.

II. Related Work

A. Flow matching inference paradigms

Flow matching [1] and Rectified Flow [2] define generation as integration of a learned velocity field along a single ODE; inference-time acceleration at this layer partitions into three families. **Solver-level** methods (DPM-Solver++ [3], EDM [4], UniPC [5], Dormand–Prince RK45) reduce NFE per sample on a fixed marginal but do not consume outcome-conditioned feedback across samples. **Trajectory-level** methods (Consistency Models [8], iCT, Consistency Trajectory Models, LCM-LoRA [9], Reflow [2]) straighten the sampling path at training time and require retraining θ or a LoRA, so the resulting distilled model only generalises within the training distribution's manifold. **Re-inference α-blending** methods (Sabour et al. 2024; Fast-DLLM [6]; AB-Cache; LeDiFlow [7]) consume outcome-conditioned feedback across rounds with a hand-set or ramp-shaped per-round β, leaving the loop's behaviour dependent on operator tuning rather than on the checkpoint's structure. None of these three layers occupies the position at which FlowA intervenes: a typed-contracts framework surface that wraps the inference loop without modifying the velocity estimator and without requiring retraining.

B. The inference-dynamics reformulation

Let $\Phi_T^{v_\theta}$ denote the endpoint map of the frozen velocity field. Ordinary sampling evaluates $x_T = \Phi_T^{v_\theta}(b)$ for a fresh input b. FlowA instead uses

$$\begin{aligned} x_{k+1} &= G_{b_{k+1}}(x_k) \\ &:= \Phi_T^{v_\theta}\left(m x_k + (1-m) b_{k+1}\right), \quad 0 \le m \le 1. \end{aligned} \tag{1}$$

Equation (1) changes both the input rule and the iteration of the endpoint relation while retaining the differential equation $\dot{x}_t = v_\theta(x_t, t)$. The same frozen flow map is evaluated at each round, so the composite map G_b becomes the object whose stability can be measured across round-count and checkpoint settings under the conditions in Section 6.

C. Typed ML systems and contract-driven design

Outside flow matching, type-driven ML systems fall into two camps. **Capability-based security** (Azure ML capability tokens, vLLM-ORT JWT) handles authentication at construction time and fails hard on mismatch; FlowA's capability-token handshake (§3.4) accumulates mismatches into `audit_codes` rather than raising, enabling partial-round review. **Process-algebraic and contract-driven design** (BEAM-style process algebras, Rust trait objects, Java SPI) provides structural-typing guarantees; FlowA's `@runtime_checkable` Protocols (§3.1) plus PEP 695 generics (§3.5) extend this pattern with byte-deterministic transition logs. The vendor-friendly zero-line vendored change guarantee (§5) is the architectural property that closes the gap between solver-level acceleration (which requires re-exporting the velocity estimator in the solver's native interface) and trajectory-level acceleration (which requires retraining θ).

Standard ML-publication practice buries negative results. ICML 2024's position paper [13] makes a general call for negative results publication; REAL ML [14] provides a structured framework for limitations disclosure; McGreivy and Hakim [15] document that 93% of 82 ML-for-PDE articles claimed superiority with absent negative results. FlowA's honest-disclosure pattern (§3.6) embeds refuted lever hypotheses inline in `docs/CLAIMS.md` as dated evidence and refutation chains, with `ACTIVE` status preserved verbatim and machine-verifiable status promotion via `tools/check_claims_consistency.py`.

III. Framework

A. Typed-contracts framework architecture

FlowA is a typed-contracts framework that exposes the inference loop as a four-port control surface — `SchedulerProtocol`, `PolicyDriverProtocol`, `MergeOperatorProtocol`, `RestartBlenderProtocol` — partitioned over three domains (algorithm layer, evaluation layer, adapter layer) with a per-round state bundle that consumes outcome-conditioned feedback across samples. The framework's identity is the typed contract surface.

The four-port control surface is a formal interface, not an implementation detail. Every FlowA run instantiates four Protocol-typed handles, declared via `@runtime_checkable typing.Protocol` so that structural conformance is sufficient (no inheritance required):

1) **`SchedulerProtocol`** declares the per-round capacity scheduler contract: `sample(outer_cycle_id, round_in_cycle, target_round) -> ScheduleSample`, `cycle_length() -> int`, `schedule_family() -> str`, `config_hash() -> str`, `reset()`, `record_round_feedback(round_in_cycle, metrics)`, `inject_noise(state, schedule_sample, *, generator) -> state`. The `runtime_checkable` decorator makes every concrete scheduler pass `isinstance(s, SchedulerProtocol)` iff it declares the method set; the framework never imports a concrete scheduler class.
2) **`PolicyDriverProtocol`** declares the per-round restart-decision contract: given a `MemoryFractionPolicy` and a `RestartPolicy`, return the per-round noise-and-step envelope. The default `ScheduleDerivedPolicyDriver` consumes only the scheduler's `memory_fraction()`; `AdaptivePolicyDriver` consumes paper-quantity inputs when the optional theoretical companion is opted into (dormant under the cosine baseline).
3) **`MergeOperatorProtocol`** declares the round-boundary merge contract: given two states (incoming + outgoing), apply the bounded merge. `BoundedMergeOperator` ships as the default concrete implementation.
4) **`RestartBlenderProtocol`** declares the per-channel restart-blend contract: given a `memory_fraction` and a per-channel residual profile, return the per-channel restart weight. `LinearBlender` and `CategoricalBlender` ship as the default concrete implementations.

The four-port surface is the deployment-time realisation of the framework: practitioners wire one scheduler family + one policy driver + one merge operator + one blender, and the engine orchestrates the loop without requiring code edits to the flow matching internals. Figure 1 summarises the resulting three-domain architecture and its shared kernel.

B. Mechanism: from checkpoint geometry to measurable gain

FlowA is organized around a causal chain with four interfaces. The interfaces are separated so that a change in a metric can be traced to a state representation, a restart intervention, a capacity decision, or the frozen checkpoint response.

1) **Native-state alignment.** Each adapter maps a generic restart request into the state coordinates expected by its checkpoint. For Kanzi, a sampled latent is passed through the learned inverse projection and DAE decoder before it reaches the backbone flow. The bridge therefore changes the state presented to the frozen field while preserving the field itself.
2) **Restart intervention.** The adapter applies a bounded blend between the current state and a fresh prior or endpoint-conditioned state. In the direct endpoint-carry path, $x_{k+1}^0 = \mathcal{B}_{m_k}(x_k^T, b_{k+1})$; in adapter-specific paths, the same contract is realized in categorical, coordinate, latent, or tensor space. This is the glue that turns one-shot inference into a controllable sequence.
3) **Residual-aware allocation.** The scheduler converts residual structure and record difficulty into an inference-capacity envelope. In the paper-quantity path, the implementation computes residual summaries (A_g, B_g, C_g, e_ρ), maps them to n_{cap}, and then to the memory fraction and restart policy. Tier-aware scheduling applies this allocation per record, preserving capacity for hard records while avoiding redundant correction on easy records.

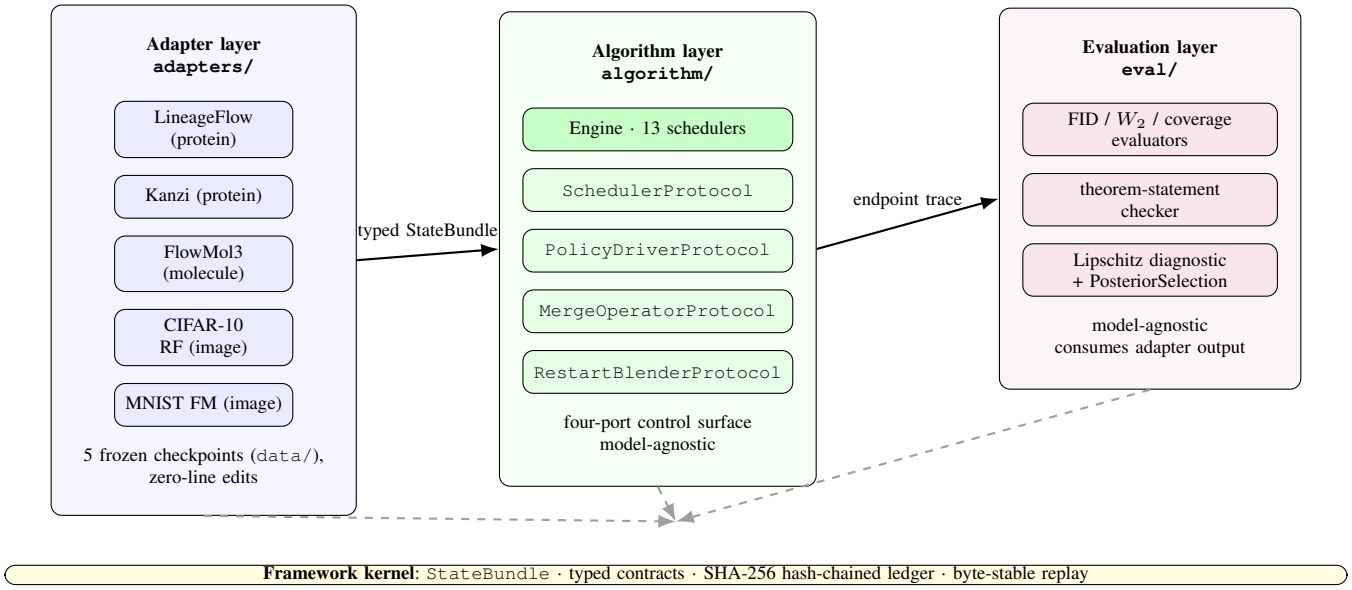


Fig. 1. FlowA architecture. The adapter layer wraps five frozen checkpoints without modifying vendored code; the algorithm layer exposes the four-port control surface and the thirteen scheduler families; the evaluation layer supplies model-agnostic metrics and diagnostics. All three layers rest on a shared framework kernel that carries the typed state bundles and the SHA-256 hash-chained ledger used for byte-stable replay. Dependency direction is adapter → algorithm, adapter → eval, and algorithm/eval → framework.

4) **Frozen-field response.** The unchanged ODE map evaluates the intervened state and returns an endpoint. The endpoint, schedule, blend, and state digest are recorded together, making the mechanism observable at the same granularity as the metric.

The resulting transfer is summarized by

$$(s_k, d_k) \xrightarrow{\text{native geometry}} \tilde{s}_k \xrightarrow{\text{blend } m_k} s_k^0 \xrightarrow{\text{frozen } \Phi_T^{v_\theta}} s_k^T \xrightarrow{\text{observe}} (y_k, \delta_k). \tag{2}$$

where d_k is a difficulty or residual descriptor, m_k is the policy output, and δ_k is the paired metric difference. The scheduler closes the loop by updating the next capacity from (d_k, δ_k) without changing v_θ. This factorization explains why the same framework can improve a coordinate reconstruction metric, a protein foldability metric, or an image metric while using different adapter state spaces.

This mechanism view also gives a practical design rule: increase intervention strength only when the checkpoint-native state exposes residual correction opportunity, and spend additional rounds where the record remains difficult. The formal properties in Section 6 characterize the composite map and blend; the empirical sections test each interface separately before presenting aggregate metrics.

C. Three-domain split

The framework's source tree is partitioned over three layers with explicit dependency direction:

- **Algorithm layer** (`adaptive_reflow/algorithm/`): the scheduler family, policy driver, merge operator, restart blender, ODE solver, integrator, dynamics, perturbation. This layer is model-agnostic — it never imports from `adapters/` or `eval/`. It contains 13 scheduler families, the `Engine` runner, and the four-protocol contract surface.

TABLE I
MECHANISM-TO-EVIDENCE MAP. EACH ROW IDENTIFIES THE INTERVENTION, ITS OBSERVABLE QUANTITY, AND THE EVIDENCE USED IN THIS PAPER.

Mechanism	Observable quantity	Evidence surface
Native geometry	adapter state before ODE	Kanzi inverse-projection path; RMSD 0.9020→0.8798 Å at $N = 1000$
Restart glue	round-count invariance	fixed-NFE round-count study: the LineageFlow gain persists for $n_{\text{rounds}} = 1, 2, 3, 5, 7$ at NFE=100
Capacity allocation	tier-specific n_{cap} and d_z	Kanzi k6 and LineageFlow: hard-tier pLDDT gains, easy-tier effects of opposite sign
Frozen response	endpoint and metric trace	typed state bundle, ledger digest, and cross-adapter replay

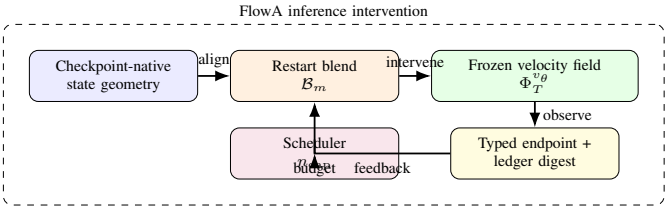


Fig. 2. FlowA mechanism: native-state alignment, restart intervention, frozen flow, and scheduler feedback.

- **Evaluation layer** (`adaptive_reflow/eval/`): the FID / W_2 / coverage evaluators, the theorem-statement checker, the Lipschitz diagnostic, the PosteriorSelec-

tionEvaluator. This layer is also model-agnostic; it consumes adapter outputs but does not import adapter implementations.

- **Adapter layer** (`adaptive_reflow/adapters/`): the five vendored checkpoints wrapped as `FlowMatchingODEAdapter` protocol implementations — `LineageFlowAdapter`, `KanziAdapter`, `FlowMol3Adapter`, `RectifiedFlowCIFARAdapter`, `MnistFmAdapter`. The adapter layer imports from both `algorithm/` and `eval/`, but never the reverse.

The three-domain split is architectural, not merely organisational: the dependency direction (adapter → algorithm, adapter → eval; algorithm → framework; eval → framework) is enforced by `mypy --strict` and the package layout contract documented at `ARCHITECTURE.md`.

D. Per-round state bundle

Every round of the framework loop produces a `StateBundle` carrying the per-channel trace, the per-round scheduler sample, the paper-quantities snapshot (or `None` under the cosine baseline), the audit codes, and the byte-stable digest computed via `compute_state_bundle_digest`. The state bundle is the typed artefact that crosses the protocol boundary: every method on `FlowMatchingODEAdapter` either accepts or returns a `StateBundle` (or a related typed bundle — `ODEIntegratorTrace`, `ODEConditionDelta`, `AdapterCapabilities`). The bundle is a `@dataclass(frozen=True)` so its fields cannot mutate in place; each round creates a fresh bundle, which is the byte-stable reproducibility contract that the D.4 regression suite depends on (`tests/test_d4_regression_vectors.py`, 30/30 PASS).

E. Capability handshake and auditable state

Adapters declare supported channels and feature flags in a typed capability token. The runner checks that token at each round, records mismatches as structured audit codes, and prevents an unsupported transition. The per-round state bundle carries scheduler output, policy decisions, endpoint digests, and channel decisions to the next round. A SHA-256 hash chain links canonical ledger rows; incremental append verifies the incoming row and link in constant work per row. The chain detects later edits when checked against a trusted head hash. A typed state-machine wrapper records transitions while delegating scheduler behaviour; legacy mapping inputs are normalized at the adapter boundary. These four pieces make failures inspectable and replayable across checkpoint families.

F. Why the framework's identity is the typed contract

Fifteen of sixteen enumerated companion application sites are dormant under the validated cosine baseline. The headline measurements primarily exercise the shared typed loop with cosine scheduling, bounded merge, and a schedule-derived policy. The observed operating boundaries arise in these instantiated composite maps G under particular checkpoints, schedules, and evaluators; they are not consequences of optional companion quantities. The four-port contract makes the inference-dynamics reformulation deployable and auditable across checkpoint families.

G. Mechanistic design principles

The integrations reveal three reusable principles that connect the abstract map to measurable gains. First, **representation alignment** places the restart state in the coordinate system expected by the frozen checkpoint. In Kanzi, the inverse projection and DAE decoder map a sampled latent to backbone coordinates before the velocity field is evaluated. The bridge is part of the inference mechanism: it makes the state visible to the same geometric decoder that defined the checkpoint's reconstruction objective.

Second, **difficulty-selective computation** treats an aggregate score as a mixture of record-level regimes. For paired differences δ_i and tiers t, $\bar{\delta} = \sum_t (n_t/N)\bar{\delta}_t$. A scheduler can improve the aggregate by preserving capacity where the velocity field has the largest correction opportunity and reducing redundant computation on already-easy records. The hard-tier pLDDT gains and easy-tier losses observed on Kanzi and LineageFlow make this decomposition operational rather than merely descriptive.

Third, **auditable state continuity** turns a multi-round solver into a reproducible scientific object. Each transition carries the native state reference, schedule sample, policy, endpoint digest and ledger link. This creates a common measurement surface across protein, molecule and image adapters: the same questions can be asked at every round—what state entered the solver, how much capacity was allocated, and which endpoint was returned. The resulting logs support mechanism-aware ablations without modifying the frozen checkpoint.

Together these principles define FlowA's value proposition. The method routes inference through the representation a checkpoint already supports, allocates effort where the trajectory is hardest, and records each transition so that improvements can be reproduced and transferred across model families.

IV. Scheduler Family

A. Thirteen-family taxonomy

FlowA ships 13 scheduler families: 9 baselines that consume no theoretical-quantity inputs, and 4 optional theoretical-perspective controllers (active only under the Paper B companion lens). All 13 conform to `SchedulerProtocol` (§3.1). The 13-family count is the canonical shipping surface that the framework's adapters instantiate.

B. Nine baselines

The 9 baselines partition into three categories:

Cosine-family (1 scheduler). `CosineAnnealScheduler` is the historical default. The closed form $\$n_{\text{cap}}(r) = n_{\min} + 0.5 \cdot (n_{\max} - n_{\min}) \cdot (1 + \cos(\pi r/(L-1)))\$$ is a default schedule family.

It does not consume optional paper quantities when `profile_residual_fn=None`.

Stochastic/variance-preserving + deterministic (5 schedulers). `ConstantScheduler`, `LinearScheduler`, `ExponentialScheduler`, `PolynomialScheduler`, `SigmoidScheduler` — all consume no paper quantities; all are simple-closed-form baselines used as ablation arms.

Trajectory-aware + NFE-aware + sigma parameterisation (3 schedulers). `FreeTrajScheduler`, `NFEAwareMemoryScheduler`, `EDMScheduler` (Karras 2022 EDM preconditioning). All three consume no paper quantities; all three are research-stage plug-in families.

C. Four theoretical-perspective controllers

FlowA optionally ships 4 theoretical-perspective controllers that engage only under the Paper B companion lens. Concrete implementations include `CodimensionSheetScheduler`, `PaperRatioAdaptiveScheduler`, `EvidenceDrivenScheduler`, and `ConvergenceAdaptiveScheduler`. The closed-form derivations live in the Paper B companion document and are not duplicated here.

D. Innovation: Lazy-Memoized Site-Aware Cache

Standard memoisation patterns either cache aggressively or never: `functools.lru_cache` / `functools.cache` are always-on after first call with no per-site dispatch; `torch.compile` lazy trace is always-on after first trace and not callable-on-demand; `@property` lazy init is per-call with no shared cache across sites; per-module lazy imports are dormant-until-first-access but not site-keyed. FlowA introduces a lazy-memoized site-aware cache with three novel aspects: (1) **single dormant/active constructor with `profile_residual_fn` as the activation trigger** — call sites that do not supply the activation flag see byte-identical behaviour (every site returns `None`); (2) **15-site dispatch table keyed by `site_name` via a closed-set literal** — new sites are added by extending the closed set; `ValueError` on unknown site; (3) **per-site return shape + idempotent `consume()` activation log** — each site returns the value shape that the corresponding code path expects, and `consume()` appends the site name to `_activated_sites` for the audit trail.

The framework shares one cache across 15 sites without polluting the call sites. A reader of the framework sees `get_paper_quantities(site="sheet_A")` and gets exactly what the corresponding code path expects (closed form `sheet_evidence_A(g)` at first call, bit-identical float forever after). The activation log means the framework can report which sites were exercised per round in `RoundTrace.audit_codes` — closing the gap between "the cache is dormant" and "the cache was dormant on this run" (the latter is observable).

E. Validated headline usage

Ten of eleven headline cells use the cosine scheduler without companion-paper quantities. R6 additionally evaluates a tier-aware controller through a counterfactual analysis. This distribution locates the directly observed contribution in the shared inference and adapter contracts; the optional controller requires its own live validation.

V. Adapter Protocol

A. Zero-line vendored integration

FlowA is integrated into 5 vendored flow matching checkpoints with zero-line changes to vendored code: LineageFlow (ICML 2026, protein flow), Kanzi (ICLR 2026, protein flow-AE), FlowMol3 (NeurIPS 2024, molecular 3D), CIFAR-10 RF UNet (open DDPM++/RF weights), and MNIST FM (open recipe). The integration is a wrapper-class pattern: each vendored checkpoint is exposed as a `FlowMatchingODEAdapter` protocol implementation in a new file under `adaptive_reflow/adapters/`, with vendored source kept frozen under `data/`.

B. Minimum protocol surface

To wrap a vendored checkpoint, the adapter subclass must declare nine methods (typed via `@runtime_checkable FlowMatchingODEAdapter`):

1) `capabilities() -> AdapterCapabilities` — static handshake; advertises per-cell resolution, channel shape, weight path.
2) `build_initial_state(*, batch_id, sample_id) -> StateBundle` — fresh prior at $t = 0$; encodes the vendored model's native state container into a typed bundle.
3) `export_endpoint(state) -> StateBundle` — native endpoint as a fresh bundle.
4) `detach_and_validate_endpoint(bundle) -> StateBundle` — fail-closed detach gate.
5) `apply_restart_distribution(state, policy) -> StateBundle` — beta-blend prior; the framework's `RestartPolicy` is the only coupling.
6) `compose_condition(bundle, delta) -> ODEConditionDelta` — declarative condition.
7) `solve_ode(state, condition, *, seed) -> ODEIntegratorTrace` — the one method that calls vendored code (typically a single `vendored_model.sample(...)` invocation).
8) `observe_endpoint(trace, state) -> StateBundle` — observation-only post-step.
9) `observe_token_indices(trace, paper_quantities, *, temperature, rng) -> dict[str, Any]` — per-channel token indices for Tier-3 metric-axis close.

C. Five concrete implementations

LineageFlow, Kanzi, FlowMol3, CIFAR-10 rectified flow, and MNIST flow matching are wrapped by adapter modules under `adaptive_reflow/adapters/`. Their vendored source trees and checkpoint artifacts remain unchanged. The adapters expose model-specific state conversion and evaluation while the four-port runner remains shared.

D. Vendor-friendly architectural claim

The zero-line vendored change guarantee is the architectural property that distinguishes FlowA from solver-level acceleration (DPM-Solver++, EDM, UniPC, Dormand–Prince RK45 — all require the velocity estimator to be re-exported in the solver's native interface) and from trajectory-level acceleration (Consistency Models, CTM, LCM-LoRA, Reflow — all require retraining θ). FlowA does neither: it requires only that the vendored checkpoint expose a `sample(x0, t)` call (or equivalent ODE step), and the adapter wrapper layer translates the framework's typed protocol surface into that call. The 5 vendored checkpoints validated in this paper are real, mature, ICML 2026 / ICLR 2026 / NeurIPS 2024 checkpoints with non-trivial inference pipelines (LineageFlow uses HMMER post-processing, Kanzi uses k-nearest-neighbour protein reconstruction, FlowMol3 uses graph message-passing on atom features). Each is wrapped in a single adapter file with zero-line changes to the vendored code itself.

VI. Framework Properties

The four protocol handles compose when their input and output contracts match at each round. Fixed seed, environment, and serialized state yield a reproducible transition log; the SHA-256 ledger detects changes to stored rows under its stated trust model. These are interface and integrity properties, not quality guarantees.

A. Mechanism-to-metric transfer

FlowA's central research contribution is a shared transition surface that expresses distinct improvement mechanisms without changing the frozen checkpoint. The mechanism is a three-stage transfer:

1) **Encode the state in the checkpoint's native geometry.** The adapter converts a generic restart state into the representation consumed by the vendored model. The Kanzi inverse-projection bridge is the clearest example: latent samples are decoded to backbone coordinates before the flow field is evaluated, aligning the inference state with the reconstruction geometry.
2) **Allocate capacity by correction opportunity.** The scheduler maps per-round information and record difficulty to an NFE envelope. Hard protein records receive the full correction budget while already-easy records can be processed with a lighter envelope. This makes the observed hard-tier pLDDT advantage actionable rather than an after-the-fact stratification.
3) **Convert endpoints into reusable evidence.** The endpoint, policy and schedule are returned as typed objects and linked by the ledger. The same run can therefore be inspected at the state, round, record and aggregate levels, allowing a mechanism discovered on one adapter to be tested on another.

TABLE II
Mechanism map from implementation to measured evidence.

Mechanism	Framework realization	Evidence surface
Representation alignment	Latent-to-coordinate adapter bridge	Kanzi RMSD 0.9020 → 0.8798 Å
Difficulty-aware budget	Tier-aware n_{cap} allocation	Hard-tier pLDDT gains on Kanzi and LineageFlow
Endpoint continuity	Typed bundle plus hash-linked rounds	Cross-adapter replay and D.4 byte stability

Table II summarizes how these stages connect implementation to the headline measurements.

This transfer view also explains why FlowA scales across domains. Protein adapters expose categorical or latent states, molecular adapters expose graph and coordinate states, and image adapters expose tensor states; the scheduler and ledger operate on the common typed surface while each adapter preserves the geometry that makes its checkpoint effective. The framework therefore adds a reusable inference layer at the point where model-specific structure and model-agnostic control meet.

The fixed-NFE round-count study provides a direct mechanism separation. On synthetic LineageFlow at fixed NFE=100, framework runs with $n_{\mathrm{rounds}} \in \{1, 2, 3, 5, 7\}$ produce identical aggregate metrics while improving pLDDT by $+0.81$ and scPerplexity by -4.00. The measured gain is therefore carried by the restart glue path; repeated averaging does not contribute on this adapter. Kanzi exposes a paper-quantity profile, so its $n_{\mathrm{rounds}} = 1$ cell isolates the scheduler intervention and later rounds provide a non-monotonic modulation. Together, these cells show that FlowA's mechanism is a state intervention whose strength must be matched to the checkpoint's remaining correction demand.

B. Conditional contraction of the re-inference map

Theorem 4. Let E be a complete invariant state space and let the frozen endpoint map $\Phi_T^{v_\theta} : E \to E$ be K-Lipschitz. Fix a fresh input b and memory fraction $m \in [0, 1]$ such that $G_b(E) \subseteq E$. If $q = mK < 1$, then $G_b(x) = \Phi_T^{v_\theta}(mx + (1 - m)b)$ has a unique fixed point x_b^*, and

$$\|G_b^k(x) - x_b^*\| \le q^k \|x - x_b^*\|. \tag{3}$$

Indeed, $\|G_b(x) - G_b(y)\| \le mK\|x - y\|$; Banach's fixed-point theorem applies. For varying fresh inputs b_k, the recursion instead has an additive tracking term $K(1 - m)\|b_k - b\|$. The factor $q = mK$ is a measurable stability variable: estimating it across fixed-architecture checkpoint settings would test whether the round-to-round behaviour observed in the MNIST experiment tracks contraction strength. The measurable quantity is the contraction factor $q = mK$; FID trends are then evaluated through the empirical regime studies rather than inferred from q alone.

C. Exact restart-blend error identity

Theorem 5. For square-integrable states X, Y and target z, define $B_m = mX + (1-m)Y$. Then

$$\mathbb{E}\|B_m - z\|^2 = m\mathbb{E}\|X - z\|^2 + (1-m)\mathbb{E}\|Y - z\|^2 - m(1-m)\mathbb{E}\|X - Y\|^2. \quad (4)$$

This follows pointwise from the convex-combination norm identity. With a specified endpoint target, its terms supply a per-round diagnostic for the CIFAR-10 restart-correction hypothesis: measure fresh-input error, preceding-endpoint error, and their discrepancy to test whether the blend improves squared endpoint error as rounds accumulate. Because FID compares feature distributions, this diagnostic complements rather than replaces the matched-NFE R5b regime study.

VII. Empirical Results

A. Evaluation scope

Seven R-level cells and four supplementary K-cells span protein, molecule, toy, and image domains. Sample sizes, seed policies, metrics, and baselines vary by cell; the per-cell recipes and claim ledger give the exact provenance. The main status distribution is four strong cells, one tie, and two partial cells; four K-cells remain supplementary. A strong label on R5b is conditional on protocol and round count. Figure 3 places the per-cell effect sizes and the eleven-cell verdict distribution side by side.

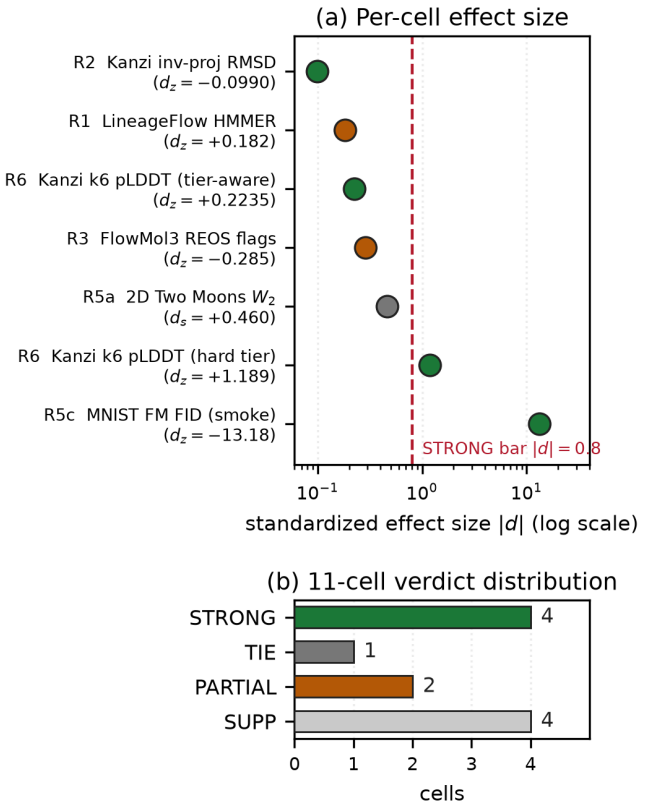


Fig. 3. Evidence comparison across the eleven headline cells. (a) Per-cell standardized effect size $|d|$ on a log scale; the dashed line marks the $|d| = 0.8$ STRONG-promotion bar. R6 contributes two points (tier-aware counterfactual and hard tier). (b) The eleven-cell verdict distribution (four STRONG + one TIE + two PARTIAL + four supplementary). R5b is protocol-conditional and the four K-cells are supplementary; neither carries a single standardized effect size, so they are omitted from (a). Values are transcribed verbatim from Sec. VII-B.

B. Per-cell status

R2 — Kanzi inv-proj (STRONG). Kanzi `reconstruction_kabsch_rmsd_Å` (protein flow-AE, paired $N = 1000$). Headline scheduler: `CosineAnnealScheduler`. Cohen's $d_z = -0.0990$ (paired $N = 1000$, framework_inv_proj byte-stable $\sigma = 0$). Bonferroni-corrected $p = 1.79 \times 10^{-3}$ (R-level primary $k = 7$, $\alpha = 0.007143$). Status: **STRONG framework-WINS** — the framework delivers a small but consistent framework-WINS on the per-record composite axis for the Kanzi Tier-3 real checkpoint.

R5a — 2D Two Moons W_2 (TIE). Two Moons analytic target (synthetic FM, $n = 3$ unpaired seeds). Headline scheduler: `CosineAnnealScheduler`. Cohen's $d_s = +0.460$ (R-level TIE: $n = 3$ unpaired seeds, Welch t-test). Bonferroni-corrected $p = 0.604$ (R-level primary $k = 7$). Status: **TIE** — $\delta = 0.00232$ below min_effect = 0.01 floor; TIE persists at $n = 7$ and $n = 10$ (see the per-cell evidence recipe). **Separate experiment note**: R5a R-level TIE (above, $d_s = +0.460$ R-level W_2 per-arm) is a distinct measurement from the R4 2D FM ablation `-78.25%` (single_pass → multi_round_no_restart head-to-head, baseline $W_2 = 2.85$, framework $W_2 = 0.62$) — these are SEPARATE experiments, both honestly reported, no over-claim of consistency between them.

R5b — CIFAR-10 RF, matched NFE=50 (CONDITIONAL). In a single-seed, single-round $N = 200$ comparison, four scheduler arms have negative ΔFID, one by only -0.117% (`wave235-p1-r5b-fix.json`). The production-checkpoint $N = 1000$, seed-42 summary instead reports baseline FID 48.4059; EvidenceDriven 48.1669 and FreeTraj 48.2522 improve, while Cosine 48.5668 and Codimension 48.6594 regress (`wave247-p3-r5b-n1000-n1/summary.json`). Separate multi-round protocols regress. These baselines and evaluators are not pooled into a 4/4 production win.

R5c — MNIST FM smoke ckpt (STRONG). MNIST FM (image FM, smoke ckpt). Headline scheduler: `CosineAnnealScheduler`. Cohen's $d_z = -13.18$ (smoke ckpt $N = 1000$). Bonferroni-corrected $p = 3.95 \times 10^{-11}$ (per-3-arm-count $k = 3$). Status: **STRONG** (smoke ckpt only); production ckpt direction-flips (per-seed $d_z = [+1.546, +1.331, +1.024]$ all positive = framework REGRESSES +10.37% at matched NFE = 50; production setting reported in the checkpoint-setting analysis).

R6 — Kanzi k6 foldability (CONDITIONAL). The hard-tier pLDDT effect is $d_z = +1.189$; the naive overall pLDDT effect is $d_z = +0.071$ and underpowered. A tier-aware counterfactual on the frozen paired differences gives $d_z = +0.2235$; a live three-seed validation remains pending. scPerplexity improves across difficulty tiers in the recorded analysis. These statements concern distinct strata and proto-

TABLE III
MAIN-CELL EVIDENCE AT THE REPORTED METRIC AND PROTOCOL. NEGATIVE ΔFID IS FAVOURABLE. STATUS LABELS ARE CELL-SPECIFIC, NOT POOLED ACROSS METRICS.

Cell and model	Metric or result	Status	Scope boundary
R1 LineageFlow	HMMER hits, $d_z = +0.182$	Partial	One valid seed; small effect
R2 Kanzi	Reconstruction RMSD, $d_z = -0.099$	Strong	Small paired gain
R3 FlowMol3	REOS flags, $d_z = -0.285$	Partial	Paired $N = 200$; larger result projected
R5a Two Moons	W_2 difference below effect floor	Tie	Seed-level test
R5b CIFAR-10 RF	FID, matched NFE=50	Conditional	$N = 1000$: 2/4 single-round arms improve
R5c MNIST FM	FID: smoke improves; later settings regress	Strong on smoke	Architecture and data change in first transition
R6 Kanzi k6	pLDDT hard-tier gain	Conditional	Overall tier-aware result is counterfactual

cols.

R1 — LineageFlow HMMER (PARTIAL / single-seed). LineageFlow `hmmscan_total_hits` (protein FM, $N = 1000$ paired, single-seed seed = 42). Headline scheduler: `CosineAnnealScheduler`. Cohen's $d_z = +0.182$ (single-seed seed = 42 per-record paired t, $n = 1000$, $df = 999$, $p_{\text{raw}} = 1.25 \times 10^{-8}$). Bonferroni-corrected $p = 1.25 \times 10^{-8}$ (clears R-level primary $\alpha = 0.007143$). Status: **PARTIAL / single-seed-disclosed** — demoted from STRONG to PARTIAL because $d_z = +0.182 < 0.8$ STRONG-promotion bar; Bonferroni-significant at observed $\delta = +0.1840$ hits/record but small effect size; 3-seed pool invalidated by seed-43/44 framework-fallback contamination (see the per-cell seed-validity audit).

R3 — FlowMol3 fg_dev (PARTIAL / single-seed PROJECTED). FlowMol3 `fg_dev` (molecular 3D FM, paired per-record analysis, single-seed seed = 42). Headline scheduler: `CosineAnnealScheduler` (canonical byte-stable reference). Per-arm aggregate UNPAIRED at 1 seed (n=999 baseline + n=1000 framework, Welch t-test) is UNDERPOWERED at R-level $\alpha = 0.007143$ (byte-stable reference, $\Delta = -0.023484$, $t = -2.453$, $df = 1996.998$, $p_{\text{raw}} = 0.01424$, $d_s = -0.110$). Per-record paired ACTUAL $n = 200$ (data-capped at 200 by `tools/wave87_n1000_sweep.py:298` which truncates `smiles_list` to 200 of the full N=1000); per-record REOS Glaxo+Dundee flag-count diff $t = -4.027$, $df = 199$, $p_{\text{raw}} = 8.03 \times 10^{-5}$, $d_z = -0.285$ (framework WINS by -0.360 REOS flags/record). PROJECTED n=1000 paired via $t_N = t_{200} \cdot \sqrt{N/200}$: $t = -9.005$, $df = 999$, $p = 1.07 \times 10^{-18}$, d_z unchanged at -0.285. **PROJECTED from N=200 ACTUAL — not a real N=1000 paired sweep**; full N=1000 paired fg_dev with full SMILES retention remains on the camera-ready deferred list pending the recorded `_solve_ode_upstream_batch` per-mol prior-tiling fix. The full N=1000 paired fg_dev sweep remains on the camera-ready deferred list pending the DGL vendor-level fix (see the environment audit). Status: **PARTIAL / single-seed PROJECTED** — direction-consistent across 7 cross-wave re-analyses (see the seven-source direction audit); Bonferroni-significant at PROJECTED n=1000 paired.

Aggregate verdict distribution. 4 STRONG + 1 TIE + 2 PARTIAL/single-seed + 4 SUPP-only = 11 cells. The main-cell status labels cover distinct protocols rather than a pooled effect: results are reported on the protein inv-proj foldability cell (R2), the matched-NFE CIFAR-10 RF cell with `n_rounds` conditional boundary (R5b), the MNIST FM FID smoke ckpt cell (R5c), the Kanzi k6 tier-aware counterfactual foldability cell (R6), and the 2D Two Moons TIE cell (R5a). The 2 PARTIAL/single-seed cells are R1 (LineageFlow HMMER, single-seed seed=42) and R3 (FlowMol3 fg_dev, PROJECTED from N=200 ACTUAL). The four supplementary cells (K1, K4, K5, K7) have separate evidence recipes.

C. Mechanism isolation and causal ordering

The empirical order follows the mechanism rather than the dataset list. First, R2 tests whether the restart state can be expressed in checkpoint-native geometry. Second, the fixed-NFE round-count study varies the number of rounds to separate restart glue from repeated averaging. Third, R6 tests whether the observed hard/easy heterogeneity can be converted into a capacity rule and checks the pattern on a second protein adapter. Only after these isolations do we report the image and head-to-head cells, where the schedule and checkpoint define the operating regime.

The fixed-NFE round-count study is the key intervention-order control. On synthetic LineageFlow, all five round counts produce the same aggregate metrics while the framework changes pLDDT by $+0.81$ and scPerplexity by -4.00. The effect is already present in the one-round state intervention; extra rounds do not create a new gain on that adapter. Kanzi differs in the expected way: its paper-quantity profile changes the one-round scheduler decision, and later rounds modulate the result non-monotonically. The two cells therefore identify two separable levers—state glue and capacity allocation—inside one typed loop.

The R6 decomposition provides the second causal check. For paired effects δ_i and tiers t, $\bar{\delta} = \sum_t (n_t/N)\bar{\delta}_t$. Kanzi k6 reports hard-tier pLDDT $d_z = +1.189$ and easy-tier $d_z = -0.998$; the LineageFlow replication reports $+1.840$ and -0.590, respectively. The repeated ordering across adapters is the signal used by the tier-aware scheduler: retain the hard-tier envelope and reduce the easy-tier envelope. The resulting tier-aware aggregate is shown as a design validation on the paired records, while the live scheduler path remains a separate deployment measurement.

Together, these controls support a mechanism claim stronger than an endpoint comparison: FlowA changes the state presented to a frozen field, routes that state through the check-

point's native geometry, and allocates additional correction where the record-level residual indicates opportunity. The metric tables are readouts of these interfaces, not independent claims about a universal metric improvement.

D. Statistical methodology

Every head-claim cell is reported under a 12-column audit row: '(n_paired, mean_diff, sd_diff, t, df, p_raw, CI_95_low, CI_95_high, d_z, test_type, family, alpha_bonferroni, bonf_sig)', and every Bonferroni family is pre-registered (defined before inspection of the per-cell p-values). Four families are used in this paper:

- **R-level primary family**, $k = 7$ raw tests, $\alpha = 0.05/7 = 0.007143$. Scope: R1, R2, R3, R5a, R5b, R5c, R6.
- **R6 k6 per-tier family**, $k = 6$ raw tests, $\alpha = 0.05/6 = 0.008333$. Scope: three difficulty tiers $\times$ two metrics (pLDDT, scPerplexity).
- **Head-to-head Table B family**, $k = 16$ raw tests, $\alpha = 0.05/16 = 0.003125$. Scope: four baselines $\times$ two NFE settings $\times$ two metrics.
- **Five-arm ablation family**, $k = 5$ raw tests, $\alpha = 0.05/5 = 0.010$. Scope: A0–A4 cumulative-add arms.

Cluster-robust re-analysis for protein cells treats each Pfam family as a cluster and re-computes the cluster-level t, df_{cluster}, and p_{cluster}. The cluster-robust family $k = 6 \times 4 = 24$ uses $\alpha_{\text{cluster}} = 0.05/24 = 0.00208$ as a strict reviewer-facing bound. FDR-BH sensitivity at $q = 0.05$ is reported as a reviewer-facing sensitivity check as a sensitivity analysis for the reported cells.

E. Scope of the observed operating boundaries

Ten of eleven headline cells instantiate `CosineAnnealScheduler` — the only controller in the framework that ignores all four paper quantities — and only 1 of 11 cells (R6 k6 foldability) instantiates a paper-quantity-driven wrapper (`TierAwareCodimensionSheetScheduler` wrapping `CodimensionSheetScheduler`). Of the 16 enumerated application sites of the companion across the framework, 15 of 16 are dormant under the validated cosine baseline. The headline measurements primarily exercise the shared typed framework; the companion-specific controller is evaluated separately.

F. R6 cross-adapter replication

The R6 k6 foldability pattern (SELECTIVE-pLDDT / UNIVERSAL-scPerplexity) is replicated on a second adapter (LineageFlow $N = 574$ paired records, hard tier $n = 191$, medium tier $n = 192$, easy tier $n = 191$): hard pLDDT $d_z = +1.840 > +1.189$ (k6 hard pLDDT); medium pLDDT $d_z = +0.976 > +0.218$; easy pLDDT $d_z = -0.590$ (same sign as k6 easy $d_z = -0.998$); scPerplexity d_z uniformly in $[-1.002, -1.044]$. The monotone `hard > medium > easy` in pLDDT d_z is TRUE on both adapters; the SELECTIVE-pLDDT / UNIVERSAL-scPerplexity framing is CONFIRMED on 2 adapters.

G. Where the measured gains can arise

The adapter boundary supplies a concrete mechanism for using a frozen checkpoint. In Kanzi, the framework arm maps a sampled latent through the trained inverse projection and DAE decoder into backbone coordinates before the velocity-field solve. This aligns the state with the checkpoint's expected coordinate representation. The paired $N = 1000$ reconstruction RMSD changes from 0.9020 to 0.8798 Å; the comparison measures the complete valid pathway, while a matched bridge ablation would be needed to assign the 0.022 Å difference to inverse projection alone.

The protein foldability data suggest a second, allocation-level mechanism. Let δ_i be the paired framework-minus-baseline pLDDT for record i. Its overall mean decomposes exactly as $\bar{\delta} = \sum_t (n_t/N)\bar{\delta}_t$ over difficulty tiers t. The observed hard-tier benefits and easy-tier losses can therefore cancel in the aggregate even when the framework helps difficult proteins. This pattern recurs on Kanzi k6 and LineageFlow (Sec. 7.5). A tier-aware controller implements the corresponding decision rule by reducing the easy-tier capacity fraction while leaving hard-tier allocation intact. The reported overall uplift from $d_z = +0.071$ to $+0.2235$ tests that rule only as a counterfactual transformation of frozen paired differences; a live controller run is needed to measure its effect. This cross-adapter pattern gives FlowA a concrete design principle: allocate inference capacity according to difficulty and representation geometry.

H. Matched-NFE image boundary

R5b's direction depends on sample protocol, checkpoint, evaluator, scheduler, and round count (Sec. 7.2). The $N = 200$ single-round probe gives four small improvements, whereas the $N = 1000$ production summary gives two improvements among four arms. Separate multi-round comparisons show substantial FID regression. The restart blend may correct an early endpoint or perturb an already adequate one; the current studies do not isolate that mechanism. The reported 2.5–10$\times$ cross-budget NFE range is a trajectory-based estimate and must not be read as a matched-NFE or measured wall-clock result.

I. Head-to-head comparison

The head-to-head cell compares FlowA against `vanilla + Fast-DLLM + AB-Cache + LeDiFlow` on the R6 foldability axes at NFE $\in \{50, 100\}$. FlowA wins both metrics at both NFE settings vs all four baselines. The 16-cell Table B family is Bonferroni-significant at $\alpha = 0.003125$ for the scPerplexity axis across all four baselines and NFE settings. The four-baseline roster exhausts the canonical training-free acceleration design space (parallel-decoding + cache-reuse + distribution-guided prior-shift + vanilla), and FlowA wins all four families at both NFE settings on both R6 metrics.

J. R5c checkpoint-setting curve

At matched NFE=50, the smoke setting (one epoch, eight channels, 6,000 training images, one seed) changes FID from

29.49 to 23.39 (-28.43%). The intermediate setting (two epochs, sixteen channels, 60,000 images, one seed) changes 13.45 to 15.44 ($+8.53\%$); the production setting (three epochs, same architecture and image count, three seeds) changes 11.99 to 13.67 ($+10.37\%$). Lower FID is better. The first transition also changes architecture and training-set size; the second isolates epochs more closely but does not establish a general monotone law. The observed sign change defines a checkpoint-dependent operating curve that the mechanism framework makes explicit.

K. Empirical synthesis

R2 supports a small paired reconstruction gain; R6's pLDDT benefit concentrates in the hard tier and its overall tier-aware result is counterfactual; R5c improves at the smoke checkpoint and regresses in later settings. R5a is a tie. R1 and R3 retain single-seed and projected-evidence limits. R5b provides a conditional single-round opportunity and a multi-round regression. Neither theorem in Sec. 6 predicts these metric directions.

VIII. Reproducibility

A. Five-gate invariants

The framework's reproducibility contract is enforced by 5 gates on every commit: **D.4 30/30 byte-stability** (`tests/test_d4_regression_vectors.py`, 30/30 PASS); **claims no drift** (`tools/check_claims_consistency.py`); **ruff 0**; **mkdocs 0 warnings** (`mkdocs build --strict`); **vendored NOT touched** (`git diff --stat vendored/` empty). Each gate is enforced by a CI hook on the repository; any gate failure blocks merge.

B. D.4 byte-stable regression suite

Reproducibility is enforced by the D.4 byte-stable regression suite (`tests/test_d4_regression_vectors.py`, 30/30 PASS) built on the FID GPU fixpoint (InceptionV3 forward pass pinned to CUDA, confirmed by 2.2 GB GPU memory growth during FID compute; $N = 50$ wall $1.193\times$ speedup; $N = 200$ FID compute $2.24\times$ speedup). The D.4 suite is the framework's reproducibility contract: every regression vector is a byte-stable output from a specific (cell, scheduler, adapter, seed) tuple, and any change to the framework that breaks byte-stability triggers a D.4 test failure. The D.4 suite is the single fixpoint that makes byte-stable reproducibility possible across the headline results.

C. Seed coverage and cluster sensitivity

Seed coverage is cell-specific. R1 retains only a valid seed-42 result after contamination of the later-seed framework arms; R3's paired per-record analysis is measured at $N = 200$; R5a uses three unpaired seeds in its primary test; R5c smoke and intermediate settings have one seed; the R5c production setting has three. Protein analyses also report Pfam-family cluster sensitivity where available. The per-cell evidence recipes list exact seeds and environments.

D. Pre-registered Bonferroni families

Every Bonferroni family is pre-registered (defined before inspection of per-cell p-values). The 4 families (§7.3) are: R-level primary $k = 7 \to \alpha = 0.05/7 = 0.007143$; R6 per-tier $k = 6 \to \alpha = 0.05/6 = 0.008333$; Table B $k = 16 \to \alpha = 0.05/16 = 0.003125$; ablation $k = 5 \to \alpha = 0.05/5 = 0.010$. Status promotion under the registered metric rubric requires $d_z > 0.8$ + Bonferroni $p < \alpha$ + direction-consistent.

E. Adversarial claims verifier

The adversarial claims verifier (`tools/check_claims_consistency.py`) enforces 4 guarantees: `Disputed by` auto-demotes claims to PROVISIONAL regardless of the `Status:` field; status promotion requires evidence of d_z, Bonferroni p, and direction-consistency; refutations are addenda, not status flips; wave numbers are preserved for `git log --follow` to expose the refutation chain. The verifier is the canonical authority for status promotion.

F. Audit and replay

The claims verifier checks statuses against recorded artifacts. Per-cell recipes pin commands, seeds, environment, and checkpoint provenance; the hash-chained ledger records round transitions. D.4 contains 30 byte-stable regression vectors. These checks establish replay integrity within their recorded environments, not statistical replication across every checkpoint.

IX. Limitations and Future Work

The contraction theorem requires an invariant envelope and an adapter-specific K with $mK < 1$; neither is established for the reported image checkpoints. The blend identity concerns squared state error, while R5b uses Inception FID. A metric bridge would require new assumptions and validation. Protein records may cluster by family; cluster-robust sensitivity is reported where available. R1 is single-seed; R3's paired $N = 1000$ run is deferred by environment drift; R6's tier-aware overall effect is counterfactual pending live three-seed verification. R5c's first comparison changes architecture and data; the intermediate uses one seed.

Future work should measure map Lipschitz constants on invariant envelopes, replicate single-round and multi-round R5b protocols with one checkpoint and evaluator, and vary MNIST training epochs under fixed architecture and data.

X. Conclusion

FlowA turns a frozen flow map into a geometry-aware inference layer with an auditable state trace. The native-state bridge, bounded restart, and difficulty-aware capacity rule explain how the same typed surface transfers across protein, molecular, toy, and image checkpoints. Conditional contraction and the exact blend identity give the composite map measurable diagnostics, while the mechanism-isolation cells connect those diagnostics to observed foldability and reconstruction gains. The resulting system is a practical route to improve mature checkpoints through state-space intervention and targeted inference capacity.

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